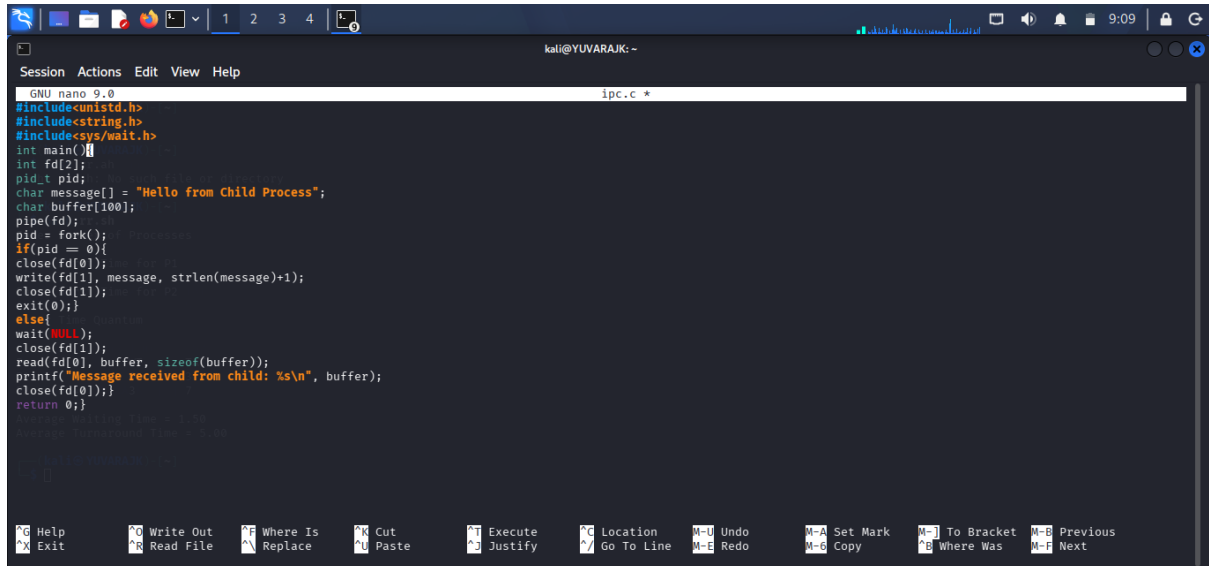
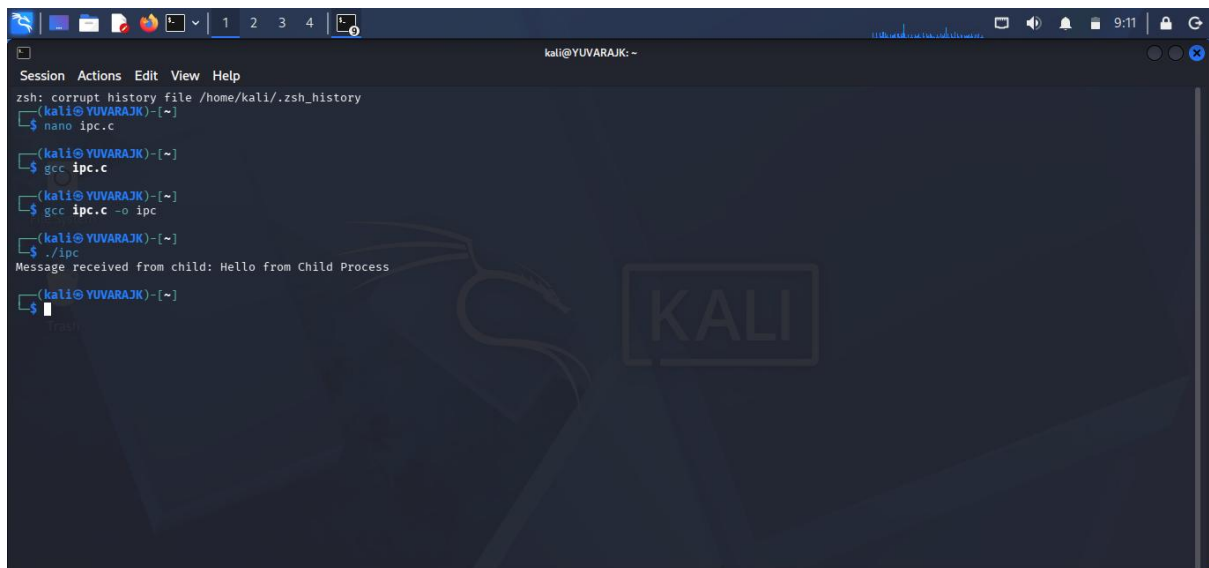


C PROGRAMMING : IPC



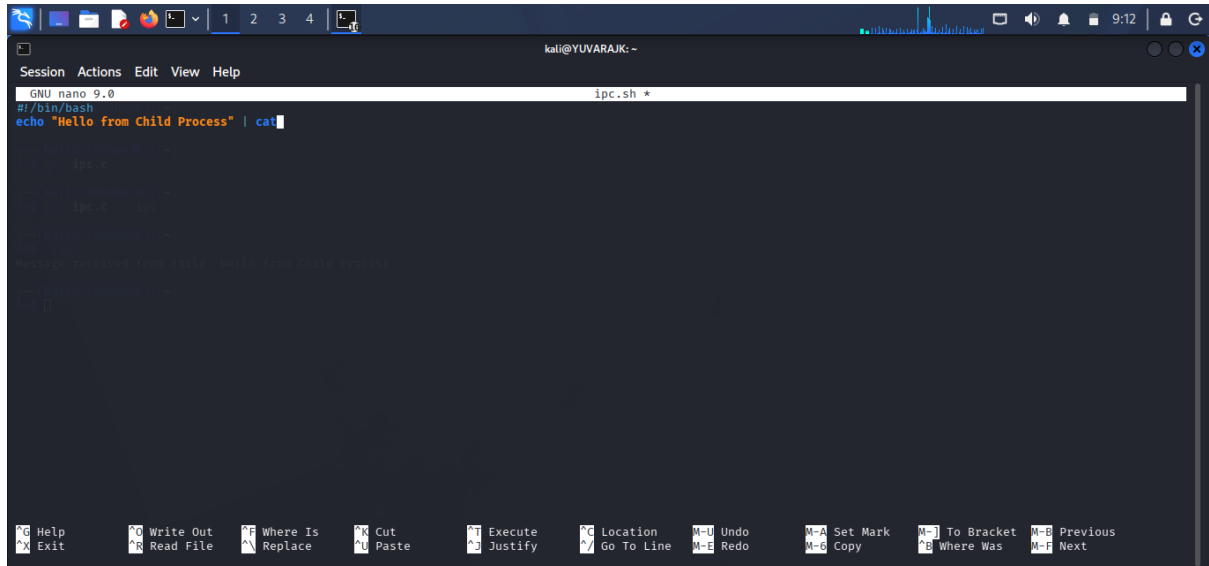
```
GNU nano 9.0 ipc.c *
#include<unistd.h>
#include<string.h>
#include<sys/wait.h>
int main()
{
    int fd[2];
    pid_t pid;
    char message[] = "Hello from Child Process";
    char buffer[100];
    pipe(fd);
    pid = fork();
    if(pid == 0){
        close(fd[0]);
        write(fd[1], message, strlen(message)+1);
        close(fd[1]);
        exit(0);
    }
    else{
        wait(NULL);
        close(fd[1]);
        read(fd[0], buffer, sizeof(buffer));
        printf("Message received from child: %s\n", buffer);
        close(fd[0]);
        return 0;
    }
}
```

OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
(kali@YUVARAJK)-[~]
$ nano ipc.c
(kali@YUVARAJK)-[~]
$ gcc ipc.c
(kali@YUVARAJK)-[~]
$ gcc ipc.c -o ipc
(kali@YUVARAJK)-[~]
$ ./ipc
Message received from child: Hello from Child Process
(kali@YUVARAJK)-[~]
$
```

SHELL PROGRAMMING :IPC



The screenshot shows a terminal window with the nano editor open, editing a file named `ipc.sh`. The script content is as follows:

```
#!/bin/bash
echo "Hello from Child Process" | cat

# Create a named pipe
mkfifo ipc.fifo

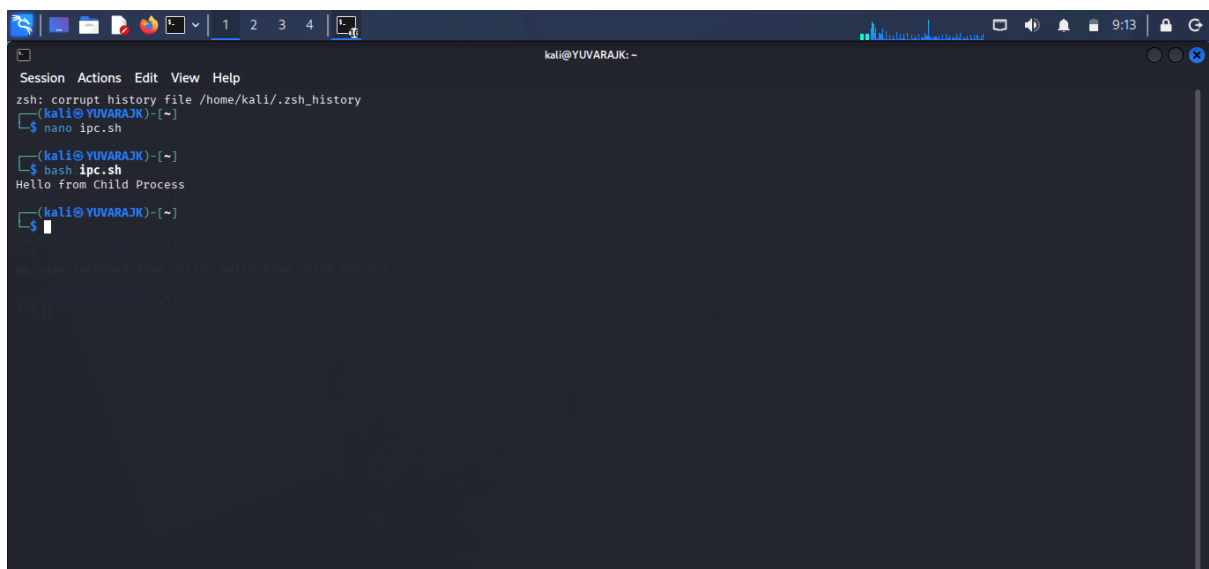
# Fork a child process
pid=$(cat /dev/null && cat ipc.fifo > /dev/null &)

# Wait for the child process to finish
wait $pid

# Receive data from the child process
read message < ipc.fifo

# Print the received message
echo "Message received from child: $message"
```

OUTPUT :



The screenshot shows a terminal window with the following commands and output:

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@YUVARAJK)-[~]
$ nano ipc.sh
(kali@YUVARAJK)-[~]
$ bash ipc.sh
Hello from Child Process
(kali@YUVARAJK)-[~]
$
```

The output of the script is "Hello from Child Process".